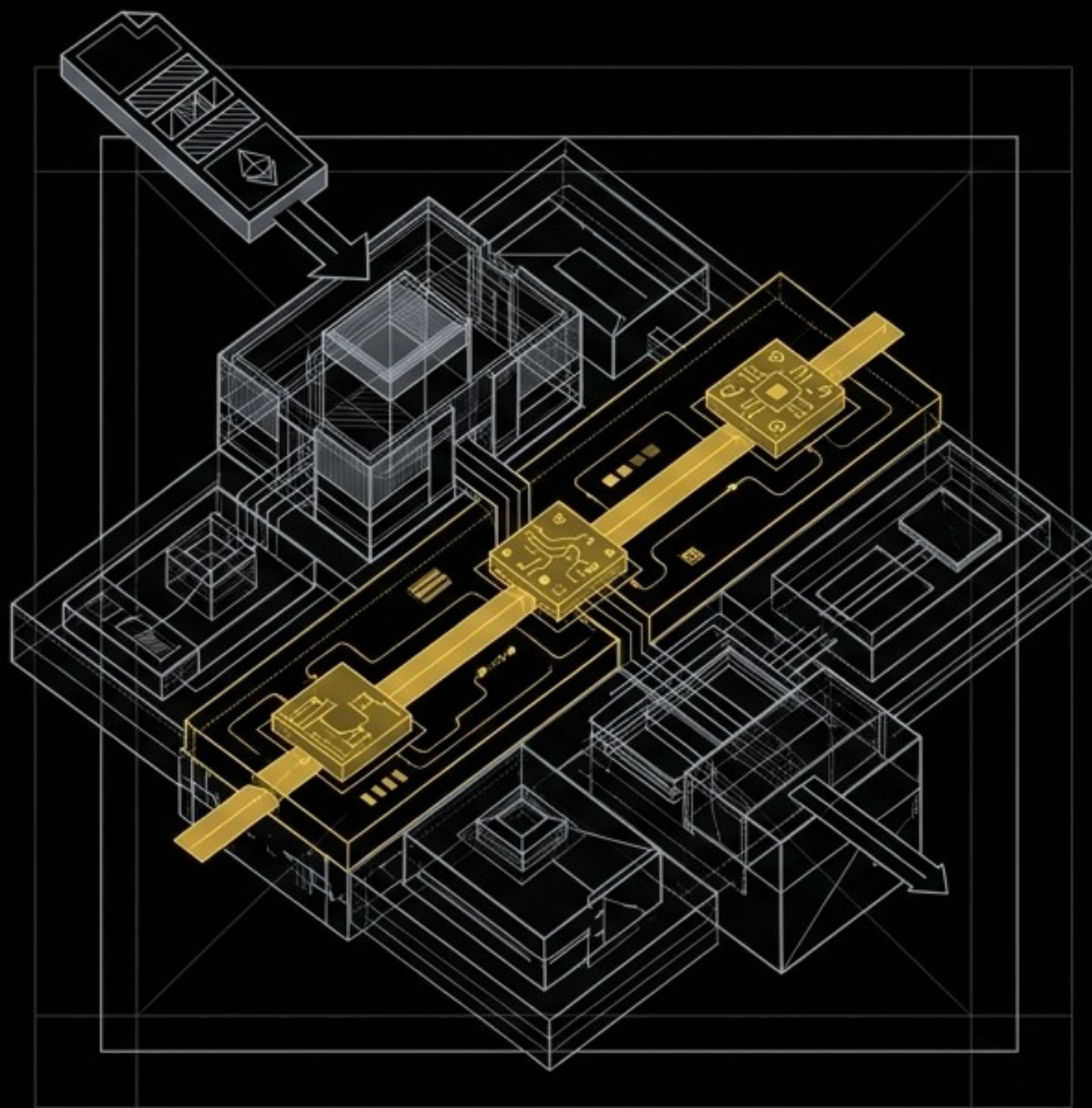




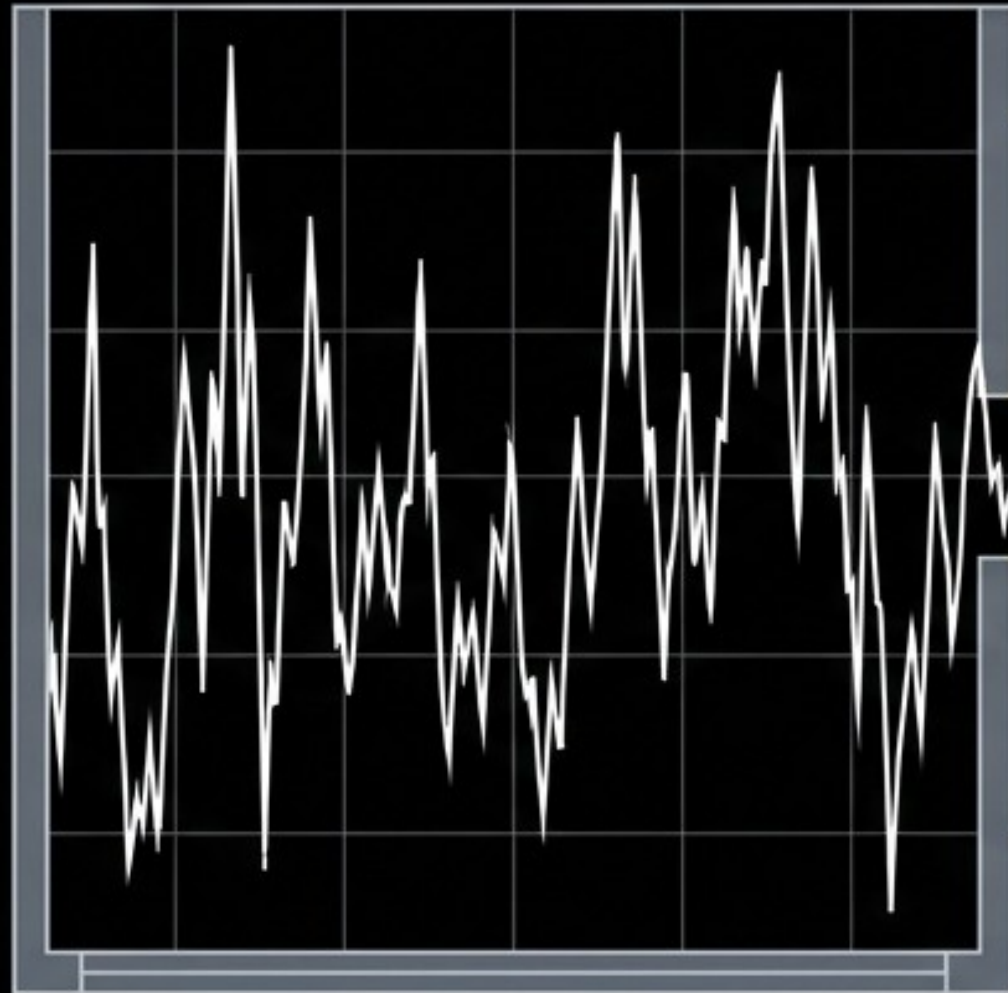
Gas Economics & Execution Halting

Mechanisms of decentralized computational markets and EIP-1559.

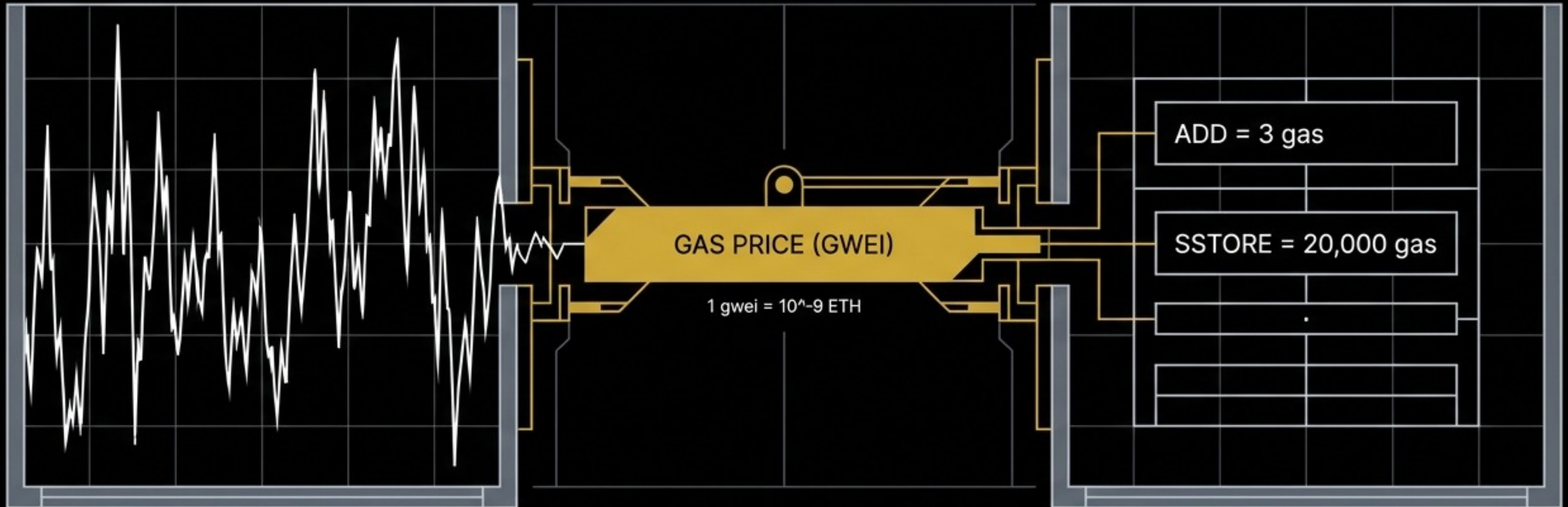
A structural deconstruction of computational scarcity, algorithmic pricing, and terminal execution states on the Ethereum Virtual Machine.

THE DECOUPLING INSULATION

EXTERNAL MARKET (ETHER)



INTERNAL EVM ENGINE (GAS UNITS)



PHYSICAL WORK VS. FINANCIAL ASSET

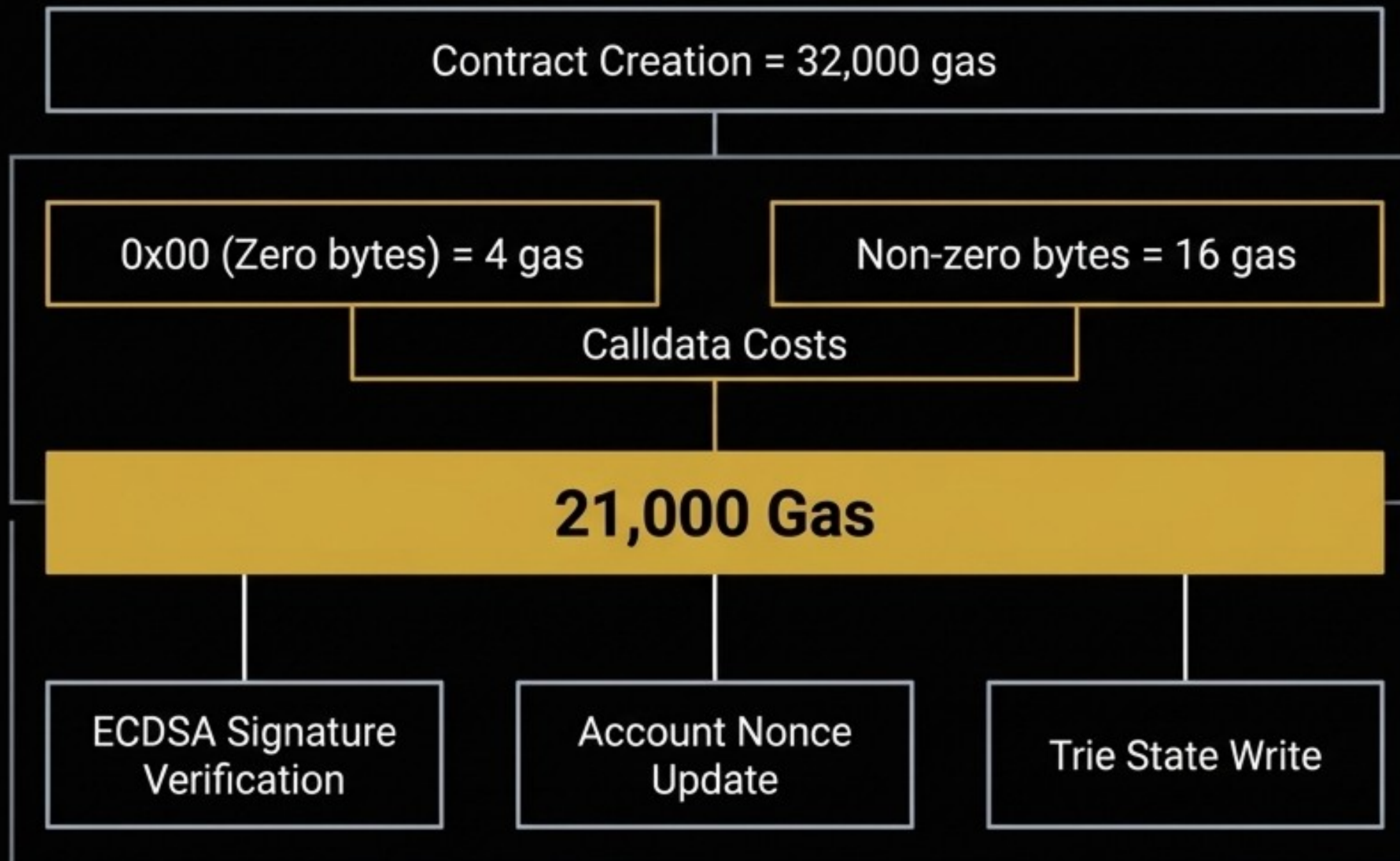
Gas measures immutable CPU and memory workloads. The physical cost of computation remains static whether ETH is \$10 or \$10,000.

THE FREE MARKET LEVER

Gas Price floats dynamically based on block space supply and demand, insulating application execution costs from crypto market speculation.

The Intrinsic Gas Floor

$$G_{\text{intrinsic}} = 21,000 + G_{\text{calldata}} + G_{\text{creation}}$$



Mempool Defense Mechanism

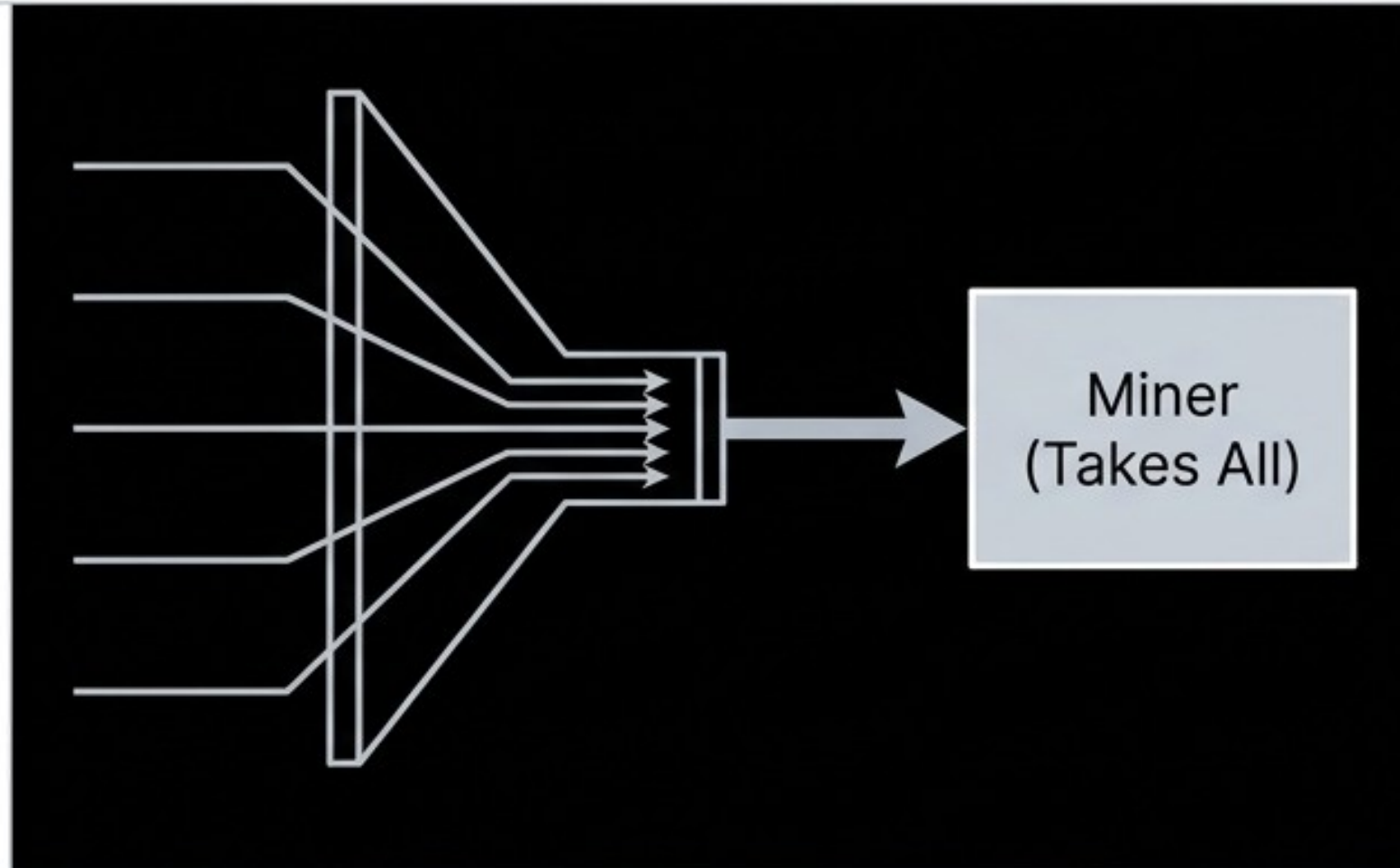
Transactions submitting a gasLimit below the $G_{\text{intrinsic}}$ floor are instantly dropped by nodes without initiating the Virtual Machine.

The Baseline

The 21,000 gas absolute minimum is non-refundable, serving as the required toll to prevent network spam and compensate for physical validator resources.

The EIP-1559 Bifurcation

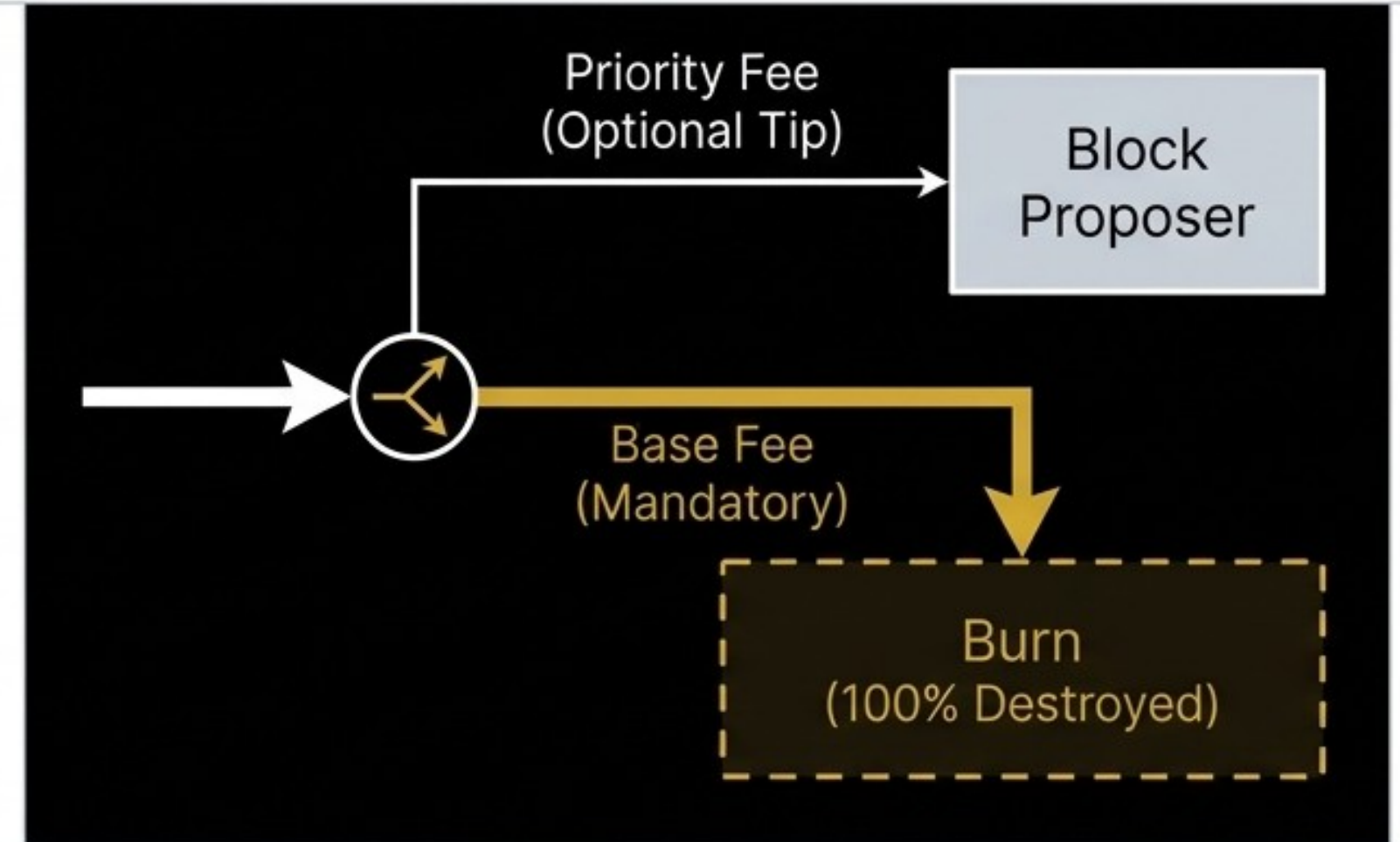
Pre-London Hard Fork



Blind First-Price Auction

- Drastic overpaying for inclusion certainty.
- Wild volatility spikes during peak network events.
- Systemic risk of miner-injected fake transactions.

EIP-1559 Architecture



Dual-Fee Architecture

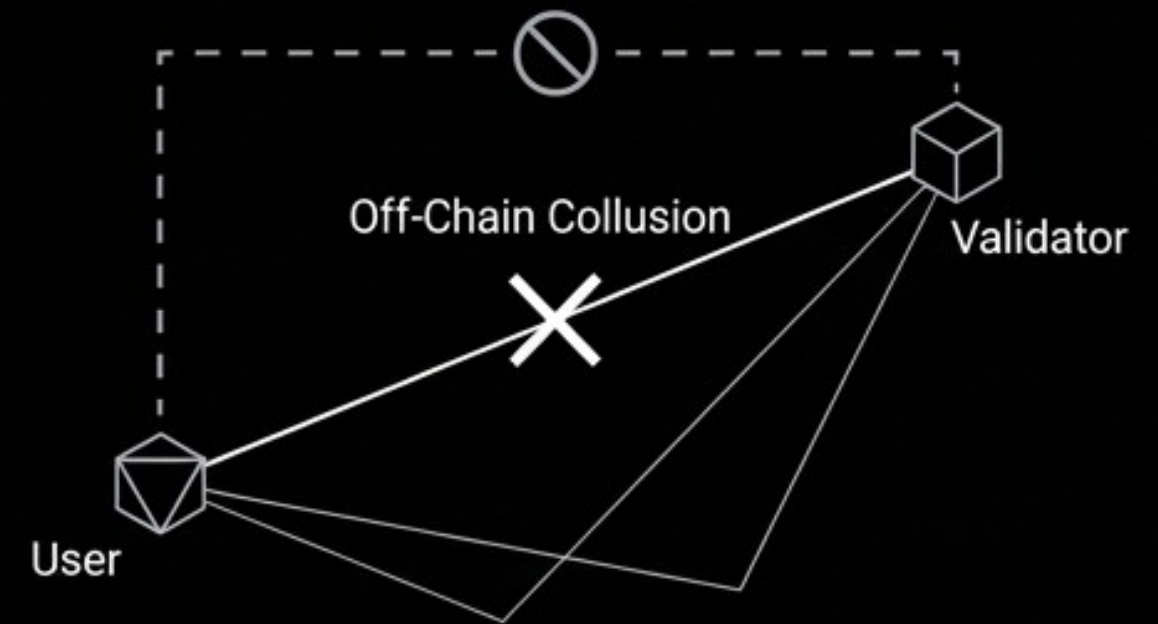
Algorithmic routing separates the mandatory protocol base cost from the optional inclusion tip, strictly governing network economics.

Base Fee Burning & Ultrasound Money

The Security Rationale

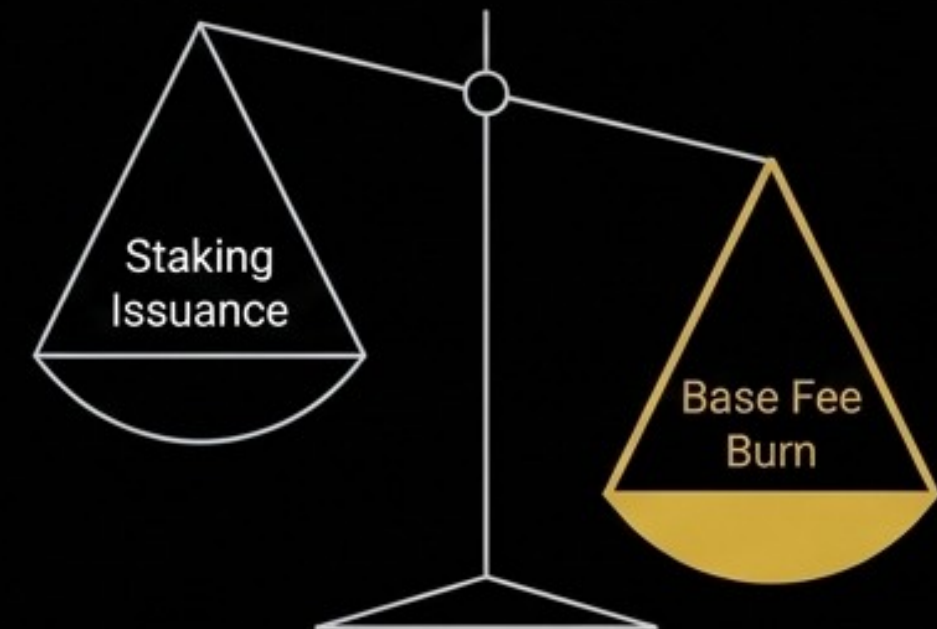
If validators received the Base Fee, they could execute off-chain side agreements to process transactions for free while artificially inflating the algorithmic base price for everyone else without economic loss.

Burning 100% of the Base Fee completely removes the financial incentive for validators to manipulate block sizes.



The Monetary Transformation

During periods of high on-chain activity, the rate of Base Fee destruction mathematically overtakes the issuance of new validator rewards. This flips the network into a deflationary state, continuously reducing the circulating supply.



The Oscillation Gauge



$$B_{n+1} = B_n * \left(1 + \frac{1}{8} \frac{(G_n - 15M)}{15M} \right)$$

Elastic Blocks

The protocol dynamically scales block capacity between 15M and 30M gas to absorb sudden traffic spikes.

Exponential Cooling

The +/- 12.5% adjustment compounds exponentially. Sustained maximum blocks (30M) force the base price up ~3.25x in just 10 blocks (2 minutes), mathematically pricing out non-urgent transactions and forcing network congestion back to the 15M equilibrium.

Effective Gas Price Calculus

User Inputs

- `gasLimit`:
Maximum compute quota
- `maxFeePerGas`:
Absolute price ceiling
- `maxPriorityFeePerGas`:
Validator tip

$$\begin{aligned} \text{EffectiveGasPrice} = & \min(\text{maxFeePerGas}, \\ & \text{BaseFee} \\ & + \\ & \text{maxPriorityFeePerGas}) \end{aligned}$$

The protocol automatically enforces the fairest possible price using the `min()` function.

Execution & Refund

$$\text{Total Cost} = \text{GasUsed} * \text{EffectiveGasPrice}$$

Origin Wallet ← Unspent gasLimit and maxFee spread automatically refunded

Execution Halting Taxonomy

1 Clean Revert

Trigger:

Explicitly initiated by business logic checks failing (e.g., `require()` detecting insufficient balance).

Data Mutation:

All state changes are atomically reversed to the pre-transaction state.

Gas Impact (Crucial):

The user is only billed for the gas consumed up to the exact point of failure. The remaining unspent gas quota is fully refunded.

2 Out-of-Gas [OOG]

Trigger:

The computational workload exhausts the user-defined `gasLimit` mid-execution.

Data Mutation:

All state changes are reversed to preserve database integrity.

Gas Impact (Fatal):

100% of the allocated `gasLimit` is seized. Zero refund is issued. This acts as a punitive measure to compensate validators for wasted CPU cycles and to deter infinite loop attacks.

The Storage Arbitrage Patch

1. Original Design

The protocol offered substantial gas refunds for clearing old storage or utilizing SELFDESTRUCT, intending to incentivize the clearing of state bloat.

2. Systemic Abuse

The Gas Token Era.

Developers created arbitrage instruments like Chi. Speculators filled dummy storage during cheap network periods to mint tokens, then burned them during congestion to unlock massive gas discount refunds, resulting in malicious block-stuffing.

3. EIP-3529 Intervention

August 2021 hard patch. Completely eliminated the SELFDESTRUCT refund. Slashed the storage clearing refund to a mere 4,800 gas. Capped the maximum allowable total refund at 20% of the transaction's total gas usage. Arbitrage loop permanently closed.

The Limits of Native Gas & The Next Frontier

Externally Owned Accounts (EOAs)

- Strict single-point-of-failure reliance on 12-word paper seed phrases.
- Catastrophic asset loss with zero recovery options.
- Users must hold native ETH merely to transfer other assets like USDC.

Smart Contract Wallets (ERC-4337)

- Unlocking programmatic capabilities via **Account Abstraction**.
- **Sponsored gas transactions** (Paymasters).
- **Biometric signing** (FaceID via Secure Enclave).
- **Social recovery** mechanisms.

Next: Module 4.4: Wallets & Account Abstraction.